

Conventions and Notations Used Throughout This Book

Our notations are fairly standard, but it may be useful to list some of them.

$\mathbf{N}$, $\mathbf{Z}$, $\mathbf{Q}$, $\mathbf{R}$, and $\mathbf{C}$ denote the sets of natural numbers (= nonnegative integers), integers, rational numbers, real numbers, and complex numbers, respectively. We always let k, m, n (possibly with accents or subscripts) range over $\mathbf{N} = \{0, 1, 2, \dots\}$.

$\dots := \dots$... is by definition equal to $\dots$;

$A \subseteq B$ A is a subset of the set B ;

$A - B$ the relative complement $\{x \in A : x \notin B\}$ of the set B in the set A ;

$f: A \rightarrow B$ f is a map from the set A into the set B .

Formally, a map $f: A \rightarrow B$ is an ordered triple, consisting of

- (i) $\Gamma(f) := \{(a, f(a)) : a \in A\}$ (the **graph** of f , a subset of $A \times B$),
- (ii) the set A (the **domain** of f),
- (iii) the set B (the **codomain** of f).

Given a map $f: A \rightarrow B$ and a set $B' \subseteq B$ with $f(A) \subseteq B'$ we indicate the map $a \mapsto f(a): A \rightarrow B'$ by $f: A \rightarrow B'$, and similarly, if $B \subseteq B''$ we indicate the map $a \mapsto f(a): A \rightarrow B''$ by $f: A \rightarrow B''$. (This may be an abuse of notation but is convenient and will not lead to confusion.) Also, with f as above and $A' \subseteq A$ we let $f|_{A'}: A' \rightarrow B$ be the restriction of f to A' . “Function” is used as a synonym for “map”. Usually we write “function” if we think of its values as numerical.

“Collection” is a synonym for “set”; we use “collection” if we think of its members as sets themselves. We do maintain the distinction between a **family** $(a_i)_{i \in I}$, which is formally the assignment $i \mapsto a_i$, and the set $\{a_i : i \in I\}$ of values of this assignment.

We let $\mathcal{P}(X)$ denote the power set of the set X , and let $|X|$ be the cardinality of the (usually finite) set X . (But we also write $|x|$ for the norm of a vector x , and $|K|$ for the polyhedron spanned by a complex K ; precise definitions of $|x|$ and $|K|$ are given in the relevant chapters. We trust that this multiple use of the notation $|\cdot|$ will not lead to confusion.)

Given a set R we let R^n be the set of all n -tuples $(x_1, \dots, x_n)$ with all $x_i \in R$. Thus R^0 contains exactly one element, the empty tuple 0 , so $R^0 = \{0\}$.

If $S \subseteq A \times B$ and $a \in A$ we put $S_a := \{b \in B : (a, b) \in S\}$, and we view S as describing the family $(S_a)_{a \in A}$ of subsets of B . Usually this situation occurs where a set R is given, and $A = R^m$, $B = R^n$, and $A \times B$ is identified with R^{m+n} .

A **partition** of a set S is a collection of *nonempty* pairwise disjoint subsets of S whose union is S . A partition $\mathcal{P}$ of S is said to be **compatible** with a subset $X \subseteq S$ (or is said to **partition** X) if X is a union of sets in $\mathcal{P}$; in particular, $\mathcal{P}$ is compatible with the empty subset of S . A **refinement** of a partition $\mathcal{P}$ of S is a partition $\mathcal{P}'$ of S that is compatible with each set in $\mathcal{P}$.

Let T be a topological space and $S \subseteq T$. If the ambient space T is clear from the context then $\text{cl}(S)$ indicates the **closure** of S in T , and $\text{int}(S)$ the **interior** of S in T . Also $\text{bd}(S) := \text{cl}(S) - \text{int}(S)$ denotes the **boundary** of S in T , and $\partial S := \text{cl}(S) - S$ denotes the **frontier** of S in T . (Some books use other terminology or notation.) If T is not clear from the context we write $\text{cl}_T(S)$ and $\text{int}_T(S)$. We say that X is an open (respectively, closed) subset of S if X is a subset of S and X is open (respectively, closed) in the induced topology on S . Given a map $f: S \rightarrow H$ from S into a Hausdorff space H and given a point $p \in \text{cl}(S \setminus \{p\})$ there is clearly at most one point $q \in H$ such that for each neighborhood V of q there is a neighborhood U of p such that $f(U \cap (S \setminus \{p\})) \subseteq V$. We write

$$\lim_{x \rightarrow p} f(x) = q$$

to indicate that q is such a point. (If in addition $p \in S$ and f is continuous at p , then this just means $f(p) = q$.)